Intelligent-Resource-Monitoring-Auto-Scaling-Pipeline

A hybrid cloud system that monitors local VM resource usage using Prometheus and automatically scales to Google Cloud Platform when CPU exceeds threshold.

Overview

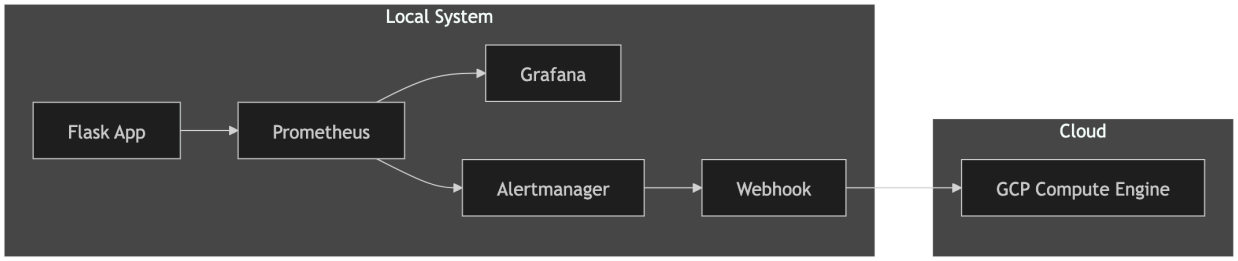
This project demonstrates a **fully automated cloud auto-scaling system** that monitors application traffic and dynamically provisions cloud infrastructure based on real-time demand.

The system uses **Prometheus for monitoring**, **Alertmanager for alert routing**, and a **custom webhook service** to trigger **Google Cloud VM creation**.

Key Features

- Real-time monitoring using Prometheus
- Visualization with Grafana
- Alerting based on traffic spikes
- Automated scaling via webhook
- Dynamic VM provisioning on GCP
- Safety mechanisms (cooldown + lock)

Architecture



Tech Stack

- **Backend**: Python (Flask)
- **Monitoring**: Prometheus
- **Visualization**: Grafana
- **Alerting**: Alertmanager
- **Cloud**: Google Cloud Platform (Compute Engine)
- **Automation**: Python Webhook + gcloud CLI

Project Structure

```
autoscaler-project/
│
├── app/                # Flask app with metrics
│   └── app.py
│
├── prometheus/
│   ├── prometheus.yml
│   └── alert.rules.yml
│
├── alertmanager/
│   └── alertmanager.yml
│
├── webhook/
│   └── app.py
│
```

```
├── grafana/
│   └── dashboard.json
│
├── start-all.sh        # Start all services
├── README.md
└── report/
```

Setup Instructions

1. Start all services

```
chmod +x start-all.sh
./start-all.sh
```

---

2. Access services

Service	URL
Flask App	http://localhost:5000
Metrics	http://localhost:5000/metrics
Prometheus	http://localhost:9090
Alerts	http://localhost:9090/alerts
Alertmanager	http://localhost:9093
Grafana	http://localhost:3000
Webhook	http://localhost:5001

---

3. Configure Grafana

- Add Prometheus datasource: `http://localhost:9090`
- Create dashboard with:

```
rate(http_requests_total[1m])
```

Testing the System

Generate Load

```
ab -n 8000 -c 200 http://localhost:5000/
```

Observe Flow

1. Traffic increases
2. Prometheus detects spike
3. Alert fires
4. Alertmanager triggers webhook
5. Webhook creates VM in GCP
6. New instance appears

Safety Features

- **Cooldown mechanism** prevents rapid scaling
- **Lock file** avoids duplicate VM creation
- Timestamp-based instance naming

GCP Integration

VM creation is triggered using:

```
gcloud compute instances create <instance-name>
```

Ensure:

- Compute Engine API is enabled
- Correct project is selected

Concepts Covered

- Observability (metrics, monitoring)
- Auto-scaling systems
- Event-driven architecture
- Cloud computing (GCP)
- DevOps automation
- Infrastructure provisioning

Final Notes

Built an end-to-end auto-scaling system using Prometheus, Alertmanager, and GCP that dynamically provisions compute instances based on real-time traffic metrics.

Author

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